

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Because E is huge, 72^{-E} is an astronomically tiny positive number, so a_2 is strictly greater than 1 but extremely close to 1.

To minimise $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$, the unique optimal solution is $x_1 = 0$, $x_2 = 1/a_2$. The infinity-norm distance from $\{0.4, 0.6\}$ to $\{0, 1/a_2\}$ is $\max(0.4, |0.6 - 1/a_2|)$.

Since $1/a_2$ is just below 1, $|0.6 - 1/a_2| = 1/a_2 - 0.6 < 0.4$. Thus the distance is exactly 0.4, which is at most 0.45.